

KEY DIMENSIONS - modelwheels.py TABLES				
ITEM (mm)	MAIN	NOSE	SOURCE	
Tyre (modelled)	8.50-18 Type III	17.5x5.25-6	0 owner decision 2026-09-26; 5 tyre makers / parts lists	
Part numbers	80606C-3 / 805-358-8	17060881 / 651-327-6	5 Goodgear / Michelin (parts listings)	
Ply / type / pressure	18 PR TL, 60 psi	8 PR TL, 60 psi	5 data book, POW placard, 556 42527	
From OD / section W	648 / 216	448 / 150	5 TRA 627-652 / 208-221; 6 photos 628-658	
Acta M - loaded R / defl.	279 / 41	287 / 15.5	6 main axle kept; 6 main 278-289, nose 280-210	
Contact patch / bleed R	268 / 199	139 / 78	6 (free choice 313 / 163)	
Section W / aspect	193.6 / 6.89	146.5 / 6.92	0 (TRA 8.90 / 6.92)	
Nose wheel / flange dia	254 / 255.2	152 / 150	5 TRA 138 207 / 6.25; 5 flange .85 / .75 in	
Between Flanges / ledge	155.8 / 34.3	127 / 22.9	5 TRA 6.25 / 5.88 in; 1.35 / 6.90 in	
Crown R / tread width	288 / 164	128 / 116	6 main, 6 nose / 6	
Shoulder R / diameter R	12 / 182	16 / 97	6 / 6 (calved tangent)	
grooves: centres +- Groove width x depth	36, 56 (2 pairs) 8 x 7	15, 44 (even) 6 x 5	6 head-on photos 3606, 3908, 3901	
Main centre x air / outer	52 / 18 / 22	24 / 23 / 11	6 outer rim to the shoulder	
wheel	split hub, 2 halves	split hub, 2 halves	5 POW 7-4-11; Goodrich 3-1543 / 3-1591	
Tie bolts in on PCD	8 on 288	4 on 110	6 hidden (main); 6 PCD, 6 count (nose)	
Web face s (clash depth)	11 (split plane)	48 (11 clamp)	6 main (covered); 6 nose (challow dish, bolts seen)	
Valve / safety plug	8.77 / 1	R 65 / 3	6 position; 5 POW	
Fuelite plug	3 (load half)	-	5 POW 7-4-11; 6 position	
Hub fairing lip / face / proud	298 / 284 / 28	-	6 3906, 3905, 3918, 3908, WDX	
fairing screws / hole	5 on 280 / 895 Ø77	-	6 5 at 72 deg	
Brake	6 Ø341.2 / 3 REF; 1 view	-	5 POW 7-4-10; 6 3-1574-1	
3 rotors / 3 stators	-	-	5 stators / 6 rotors	
Brake discs	-	-	6 tube tips / 6	
Housing lobes / rotor dia	212 / 232	-	6 (x = 8 inward)	
Brake stack s	-120 ... -28	-	6 gear.py / 6 open bore	
axle dia / bore	52 / 45	48	6 two axle (head-on photos)	
Fork axes (faces) / boss	-	+88, 118 / 83	6 3901 head-on; 3905 max	
Fork arch height / arm fore-aft	-	70 / 42-52	6 position / 6 height	
kin-protector step above flange	17 x 2.5	13 x 2.0	6 flange ring width	
Flange-face bore chamfer	8	18		

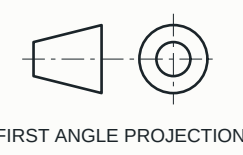
PHOTO CHECK - DEVIATION CALL-OUTS (mm)				
ITEM	PHOTOS [M]	TABLE / MODEL	Δ	ACTION
Main tyre OD (modelled)	628-658	648	8	owner decision 2026-09-26
208-210-18 (user-added)	559	-	-15	MAIN TYRE CHOICE
3-D gear MAIN TYRE	-	648	0	from wheels.py
Main section width	214-224	216	8	
Main loaded radius	279-289	284.079	4	11 flat draw
Tyre R / fairing lip R	2.15-2.21	2.16	-	
retracted tyre depth	-25 (POW 1 in)	23.5	-1.9	20-38 (rit_check 5)
Main grooves / A	6.14, 6.25	6.14, 6.26	0	3-D geometry
render groove shade	-	off (geometry)	-	done
Nose tyre OD	445	448	3	
Nose loaded radius	200-210	axle 207	0	15.5 flat draw
Nose grooves / W	6.85, 6.28	6.89, 6.30	8	
Nose fork arms	2	gear.py 2	0	3-D voke, round arch, saddle
Nose yoke arch height / half-span	6.85 (3901)	6.86 (p-3.1)	0	round arch (r2: square portal)
Nose arm width fore-aft	42-48	42-52	6	(r2: 18-92 (padle)
Nose axle nut / lock plate	gear 1 then + tear-drop	3-D both arms	0	steel, dark
Hub fairing dia / proud	-0.58 flange R	110	0	3906 nose-hub zoom
Hub fairing dia / proud	298-388 / 25-38	298 / 28	0	face > 9 W/2
Main pressure	60 psi placard	gear.py info 60	0	
Main brake	6-lobe, bright cast	3-D: brake housing	0	
Hub inboard tones	lobes > flange	housing 8.80 / wheel 9.24	0	
Loaded tread on the ground	Flat, across W	Flat (3-D)	0	r2: crown r15 only

MATERIALS (model/assembly.py NAMES)				
PART	MAIN	NOSE	NOTE	
Tyre sidewall / tread / grooves	tire / _tread / _groove	same	Matte dusty / satin / dark; NO lettering	
wheel halves	wheel_half	wheel_half	weathered cast ala (main) / gloss white (nose)	
Hub fairing	hub_fairing	hub_fairing	leg-door blue (3908), a shade darker, soft coat	
Brake housing / discs, keys	brake_housing / brake_disc	-	bright satin cast ala / heat-dark steel	
bolts, valve stem	castalum	castalum	yellow-chrome castalum / brass	
Fairing screws, valve cap	metal / black	-	rubber-sealed cap	
axle nut, hub cap	steel	-	dark steel, nut + tear-drop lock plate (nose)	
Bearing caps (nose hub ends)	chrome	chrome	polished seal rings	
arm / fork	gear_leg	gear_leg	adjacent	

LEGEND		SCALE 1:3	
	tyre (rubber) in section		0 0.05 0.1 0.25 0.2 0.25 0.38
	metal in section (hatched)		
	steel rotor / nose, shag		
	adjacent parts, free tyre, TRA env.		
	hidden		
	centre line, pitch circle		
	208-210-18 tyre (superadded)		
SCALE 1:2		0 0.025 0.05 0.075 0.1 0.125 0.158	

- NOTES
- Drawn from the parameter tables of modelwheels.py (PTable: value + source tag + note); tyre envelopes MAIN_TYRE_ENV / NOSE_TYRE_ENV, sections, rims, fairing, brake, axle; never from the mesh. Axle positions and the gear context (trailing arm, leg, shock, nose fork) from modelgear.py, drawn as adjacent parts (phantom). Photos and their measurement working stay in the ignored redscales.
 - Free-angle projection. Each assembly: SIDE view from port (geometry), FRONT view from ahead to its right (upper half in section through the axle), TOP view from above below it, for the main wheel also view C from standard (nearest) frame back at the far right. Each group side from view send on one ground line.
 - MAIN TYRE: modelled 8.50-10 Type III, OWNER DECISION 2026-09-26 tyre makers, parts lists, four photo methods: TRA 627-652, drawn free OD 648; Jane's / Pilatus drawing 208-50-10 (OD 559), superadded, shown in the accent colour. Axle and axle centre kept: axle WL 278 = the loaded radius, the tyre flat on the ground over a 290 mm patch and across the head with 141 mm deflection with 6.8 mm sidewall bulge. Nose tyre loaded too: axle WL 207 (photos 200-210), 130 mm patch, 15.5 mm deflection.
 - 3-D model gear MAIN TYRE = this envelope (R 320.0), the leg door's tyre blower (10 clear, model adding 11) and the round wet (Days WELL_L5) follow 4. modelwheels.py main_wheel / nose, wheel revolve / extrude the profiles of this sheet (tyre with the grooves, wheel halves, hub fairing, brake, valve, tie bolts) independently, 208-50-10 (AW) composes the mesh with them.
 - Tyre section: tangent axes - down R to the tread half-width, shoulder round, upper sidewall (radius solved: tangent to the shoulder, max width W/2 at 0.50 H), lower sidewall with the modelled rim-protector step (17 / 13 above the flange lip) to where the tyre leaves the rim-flange lip round. At the TRA max shoulder with the modelled section is Ø378.6, inside the TRA max Ø378.9.
 - Main hub-bearing (POW) bearings on the outer hubs, paired in the leg-door colour; no face stand 3 mm outward of the tyre's max width; retracted (wheel steel with the skin) 4 less 10.0 mm below the skin, inside the tyres 23.5 mm (POW-1, in, fit-check 5).
 - Main brake: six pistons, three retractors, steel friction surfaces, bolted to the axle (POW 7-4-10), 3 rotors / 2 stators inferred from the overload kit (2 pads, 2 rotors, 40 lobes). Housing, stack and torque tube sized inside the inboard wheel half (E); one piston tube is shown in section.
 - No markings (owner decision): no sidewall lettering, brand names or placards on wheels or tyres. The tread grooves are geometry (the render-time groove shade is off, Scribe, valve and tie-bolt clacking is the drawn pose (the wheels turn); outboard units are mirror images. The main leg door (L-D-1) is not shown.

PILATUS PC-12 PRO
WHEELS & TYRES - MAIN AND NOSE WHEEL ASSEMBLIES



DWG NO	REV	SCALE	SHEET	UNITS
PC12-L4W	A	1:3 / 1:2	A0 · 1 OF 1	mm
DRAWN				DATE
Claude - loftkit parametric model				2026-09-24

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION

